

ILES DE LA MADELEINE, QC, Canada

WMO: 717100

Lat: 47.425N

Lon: 61.775W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.6	-14.5	-20.5	0.6	-16.1	-18.2	0.8	-14.1	19.6	-7.5	18.4	-7.7	6.6	240	0.793

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	24.1	19.7	22.9	19.1	21.8	18.5	20.7	23.1	19.9	22.0	19.1	21.1	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.5	22.2	19.0	13.8	21.3	18.2	13.1	20.6	60.1	23.1	57.1	22.0	54.4	21.1	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.1	12.5	DB	-18.4	27.0	3.0	1.3	-20.6	27.9	-22.3	28.7	-23.9	29.4	-26.1	30.4	(4)
(5)			WB	-19.1	22.2	3.0	0.9	-21.2	22.9	-22.9	23.4	-24.6	23.9	-26.7	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-5.0	-6.3	-3.2	1.5	7.0	12.4	17.8	18.3	14.5	8.9	3.8	-1.1	(6)	
	DBStd	9.04	4.34	4.64	4.00	2.89	3.11	3.26	2.52	2.33	2.83	3.03	3.46	3.51	(7)	
	HDD10.0	2296	464	457	411	254	105	12	0	0	2	57	188	345	(8)	
	HDD18.3	4638	723	691	669	503	351	181	41	29	119	292	437	603	(9)	
	CDD10.0	757	0	0	0	0	12	83	242	259	136	23	2	0	(10)	
	CDD18.3	57	0	0	0	0	0	1	24	29	3	0	0	0	(11)	
	CDH23.3	74	0	0	0	0	0	3	31	38	3	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	0	1	2	0	0	0	0	(13)	
(14) Wind	WSAvg	6.4	7.7	7.0	6.8	6.4	5.7	5.4	5.1	5.2	6.0	6.7	7.2	7.8	(14)	
(15) Precipitation	PrecAvg	1049	106	77	90	80	67	66	74	77	86	101	104	115	(15)	
	PrecMax	1390	220	160	232	146	120	127	197	145	145	178	180	206	(16)	
	PrecMin	811	43	24	40	24	25	12	20	25	34	38	36	55	(17)	
	PrecStd	149	44	38	40	37	25	28	43	33	32	39	34	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	5.6	6.4	12.0	18.8	23.2	25.9	26.5	23.3	18.1	14.0	9.4	(19)	
		MCWB	7.0	5.2	5.5	9.6	13.7	18.0	21.0	21.1	19.8	15.8	13.0	8.9	(20)	
	2%	DB	4.5	3.1	4.4	9.2	15.8	20.7	24.1	24.2	21.1	16.0	12.1	7.0	(21)	
		MCWB	4.2	2.5	3.4	7.4	11.7	16.4	19.7	19.9	18.2	14.1	11.3	6.4	(22)	
	5%	DB	2.4	1.2	2.8	7.2	13.6	18.9	22.9	23.0	19.8	14.7	10.5	5.2	(23)	
		MCWB	1.8	0.7	1.8	5.6	10.4	15.4	19.1	19.2	17.3	13.1	9.6	4.5	(24)	
	10%	DB	0.8	-0.2	1.5	5.4	11.7	17.3	21.7	22.0	18.7	13.4	8.7	3.7	(25)	
		MCWB	0.2	-0.7	0.7	3.8	9.1	14.3	18.4	18.7	16.5	11.8	7.8	3.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	5.4	5.9	10.3	14.7	19.4	21.8	21.9	20.5	16.6	13.3	8.9	(27)	
		MCDB	7.0	5.5	6.2	11.3	17.8	22.0	24.7	24.6	22.5	17.7	13.8	9.2	(28)	
	2%	WB	4.3	2.6	3.6	7.8	12.6	17.4	20.8	20.9	19.0	14.6	11.5	6.4	(29)	
		MCDB	4.5	3.0	4.2	8.8	14.9	19.5	23.1	23.3	20.6	15.5	12.0	6.8	(30)	
	5%	WB	1.9	0.8	2.0	5.8	10.9	16.1	19.9	20.0	17.9	13.3	9.8	4.8	(31)	
		MCDB	2.3	1.1	2.6	7.0	13.0	18.3	22.0	22.2	19.3	14.5	10.4	5.2	(32)	
	10%	WB	0.3	-0.6	0.8	4.0	9.3	14.7	19.0	19.2	16.9	12.0	8.0	3.1	(33)	
		MCDB	0.7	-0.2	1.5	5.1	11.3	16.9	21.1	21.3	18.4	13.3	8.6	3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.0	5.2	4.7	5.9	6.0	5.6	5.2	5.0	4.5	4.3	4.2	(35)	
		MCDBR	8.4	7.8	6.6	7.6	9.1	8.4	7.0	6.4	6.1	5.8	6.7	6.7	(36)	
	5% WB	MCWBR	8.8	8.1	6.4	6.2	6.2	5.4	4.1	4.0	4.6	5.3	7.1	7.2	(37)	
		MCWBR	8.4	7.8	6.7	7.3	8.5	7.6	6.2	5.6	5.6	5.7	6.8	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.305	0.296	0.327	0.351	0.368	0.385	0.409	0.396	0.363	0.353	0.327	0.311	(40)	
		taud	2.389	2.305	2.286	2.421	2.446	2.389	2.389	2.427	2.485	2.453	2.438	2.388	(41)	
	Edn at Noon		774	867	891	903	896	879	851	848	847	795	741	716	(42)	
		Edh at Noon	70	92	110	107	109	116	114	106	91	81	67	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.89	1.79	3.20	4.36	5.27	5.64	5.59	4.96	3.78	2.31	1.11	0.68	(44)	
	RadStd		0.08	0.19	0.27	0.36	0.43	0.52	0.45	0.34	0.31	0.20	0.12	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	+14	(46)
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page